

EXERCISE 5.2
Hypothesis Testing for 1 sample

[WEEK 9]



PROBABILITY & STATISTICAL DATA ANALYSIS

Exercise #1

A principal at a certain school claims that the students in his school are above average intelligence. A random sample of thirty students IQ scores have a mean score of 112. Is there sufficient evidence to support the principal's claim? The mean population IQ is 100 with a standard deviation of 15. Assume default significance level $\alpha=0.05$.

Exercise #2

A 2023 university report states that undergraduate students typically study an average of 20 hours per week. To verify this claim, a student newspaper conducted a survey among a random sample of 10 students. The survey found that students studied an average of 17.8 hours per week, with a sample standard deviation of 3.2 hours. Does this sample provide sufficient evidence at the 0.05 significance level to reject the claim that students study 20 hours a week?